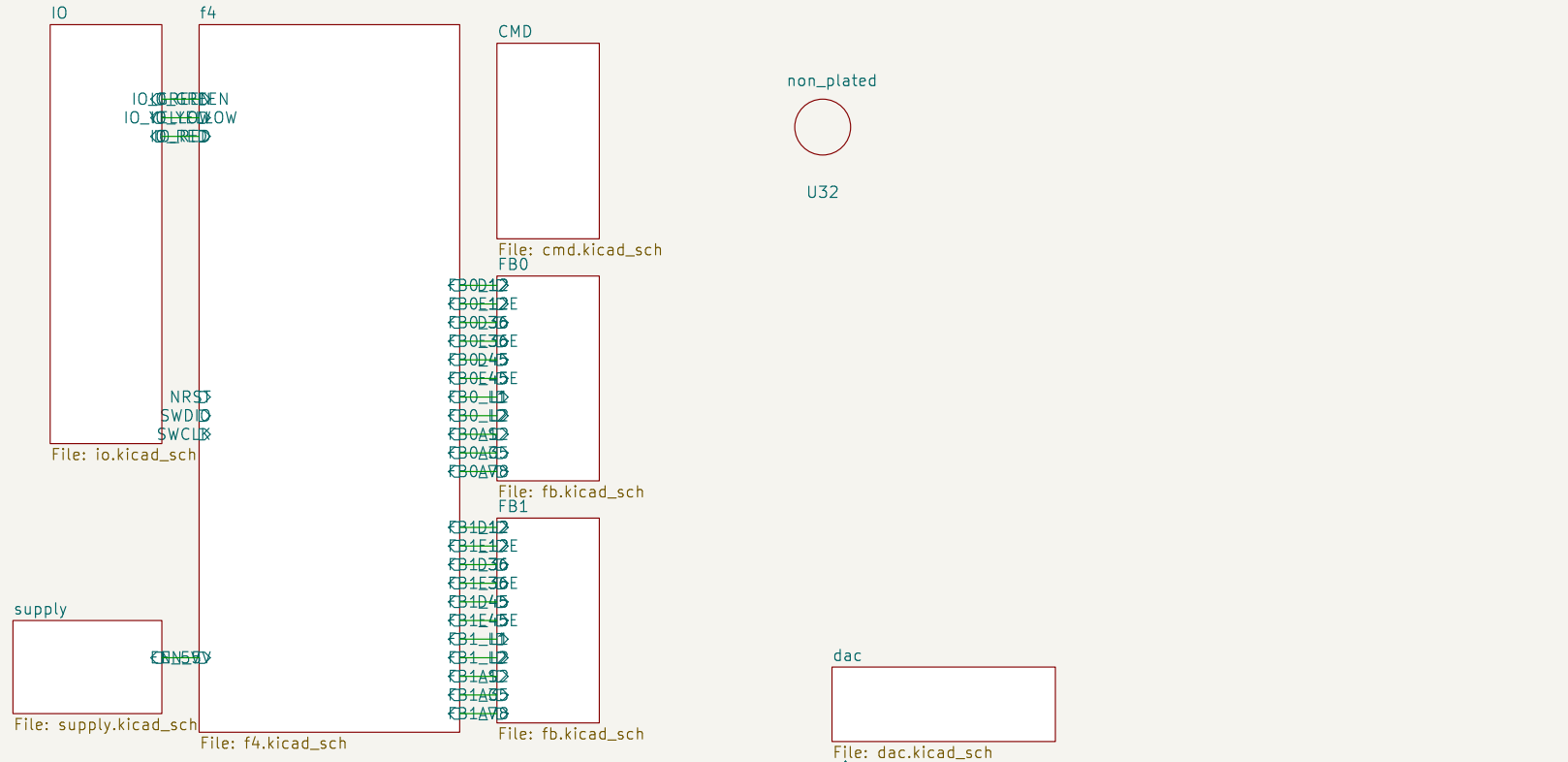
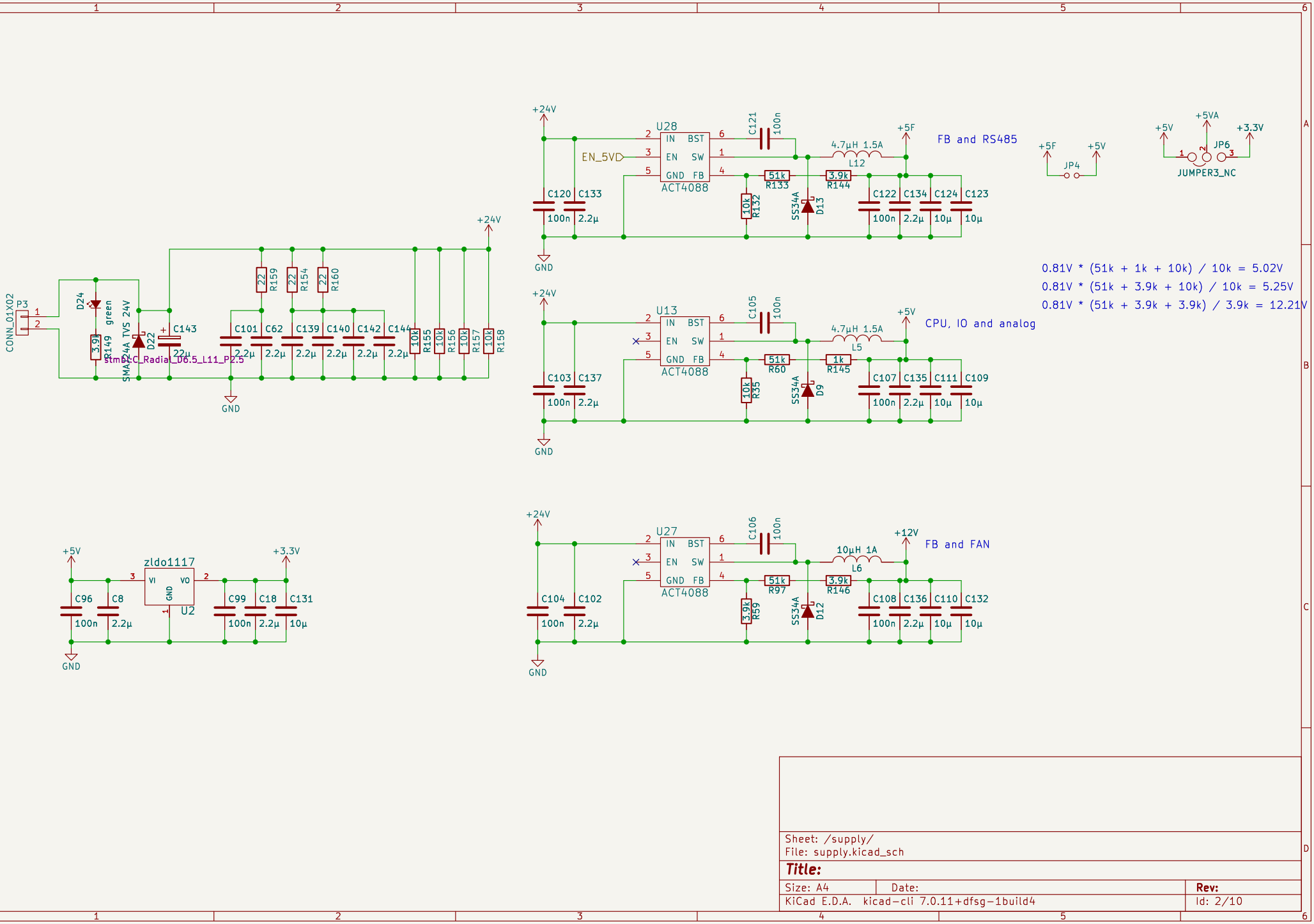


stman (docs/stman.md): stmbl v4.0's LV sheets, edited in place;
dac, ain, drvio: stman's new circuits (global labels to the MCU sheet).

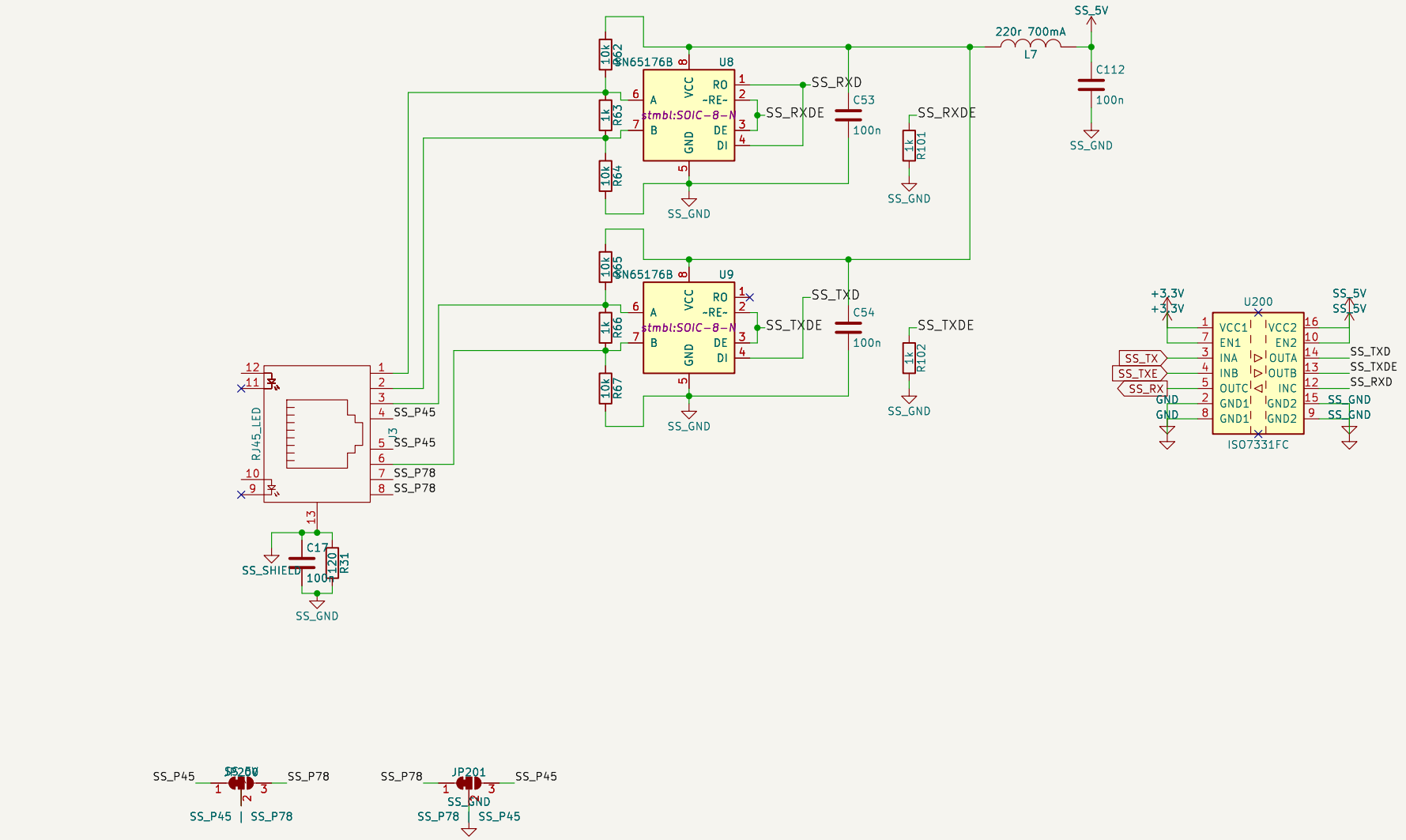
Sheet: /
File: stman.kicad_sch

Title: stman: stmbl with an analog output

Size: A4	Date:	Rev:
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/10

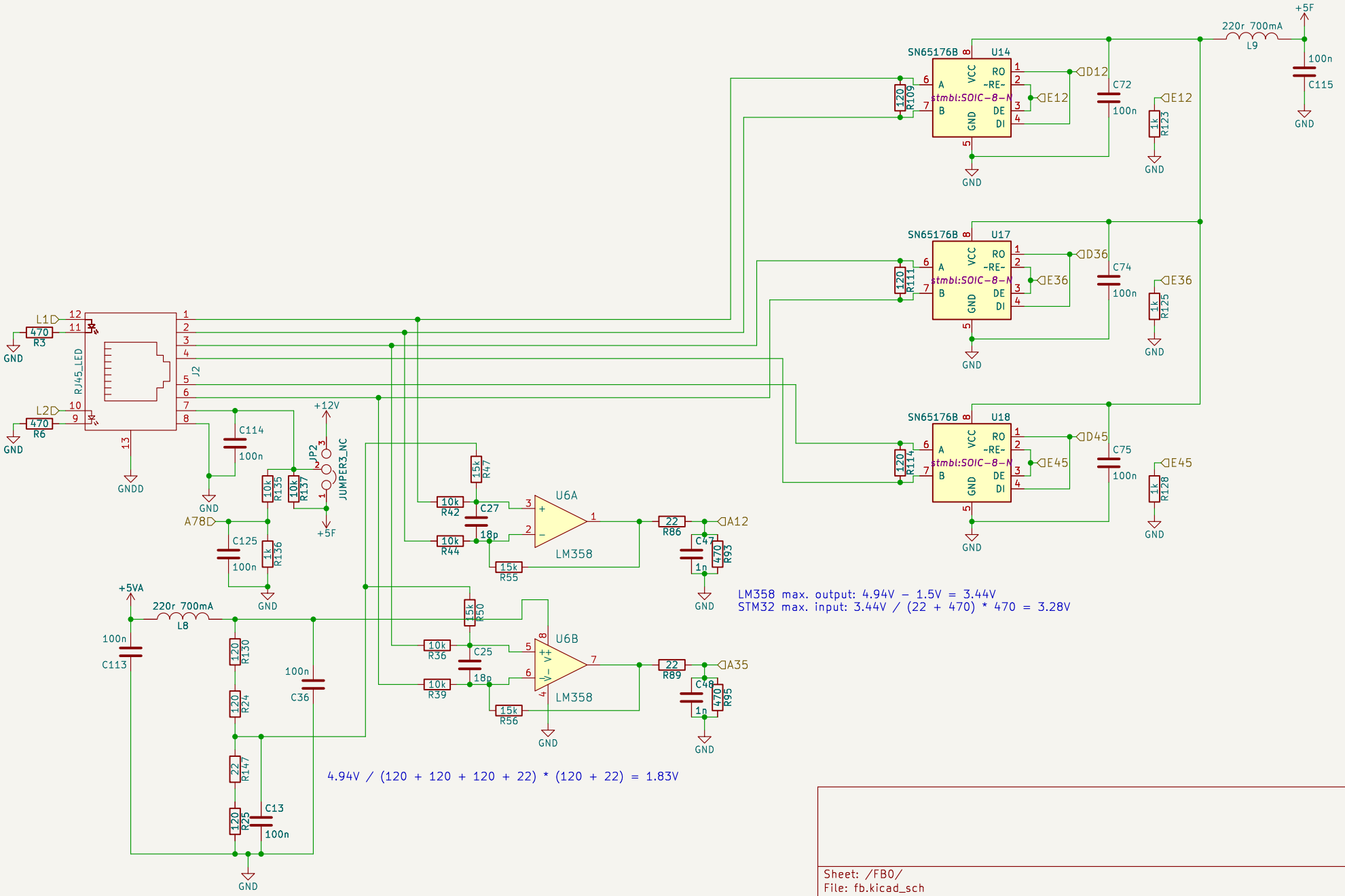


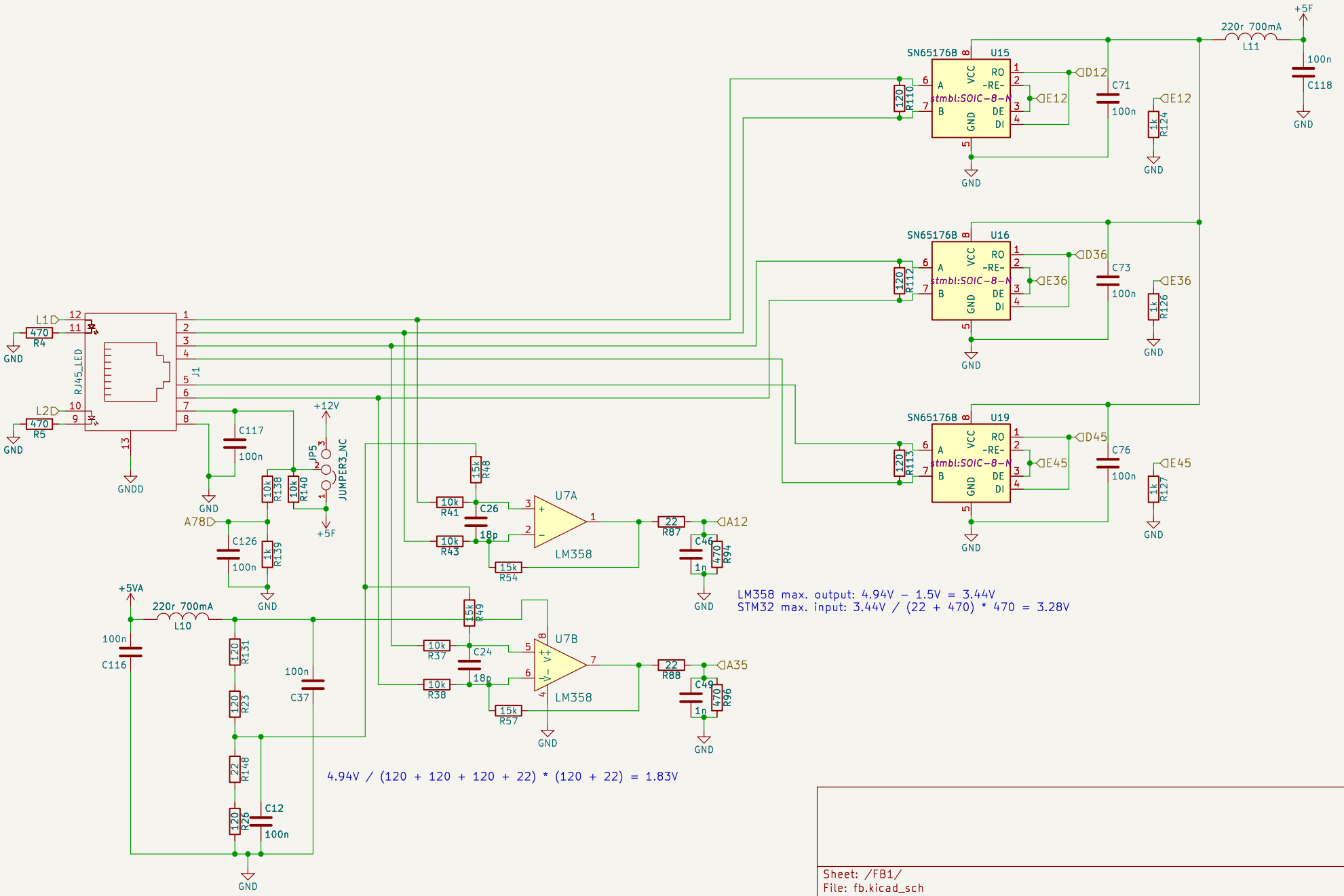
Sheet: /supply/		
File: supply.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 2/10

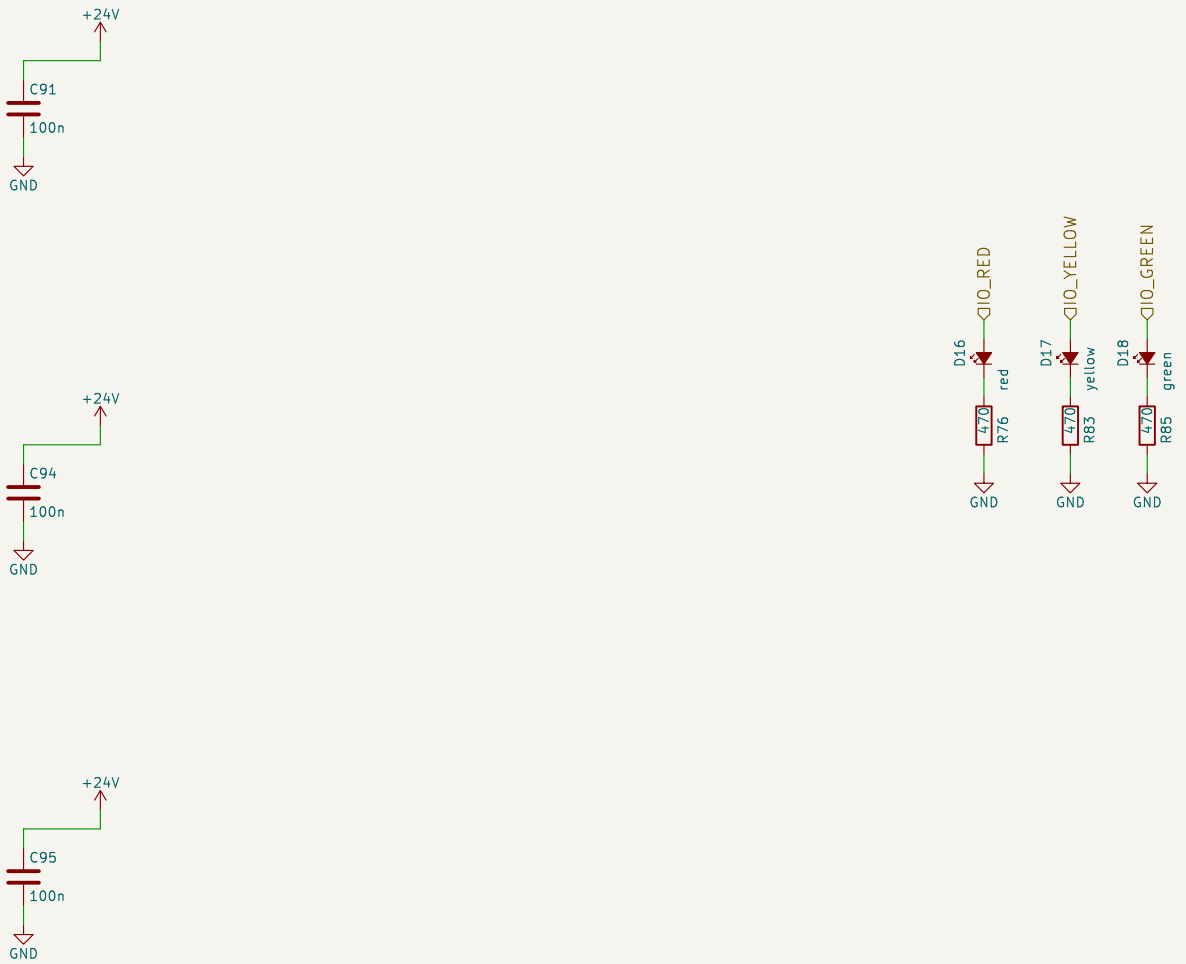


stman: the transceivers U8 (receive, pair 1/2) and U9 (send, pair 3/6) are on the cable's side,
supplied from the cable (SS_5V, SS_GND); ISO7331FC U200 (outputs low while unpowered) is the only path to the MCU.
U10, U11 (pairs 4/5, 7/8) removed: those pairs carry the cable's +5 V and GND (JP200, JP201: check against the Mesa card *)

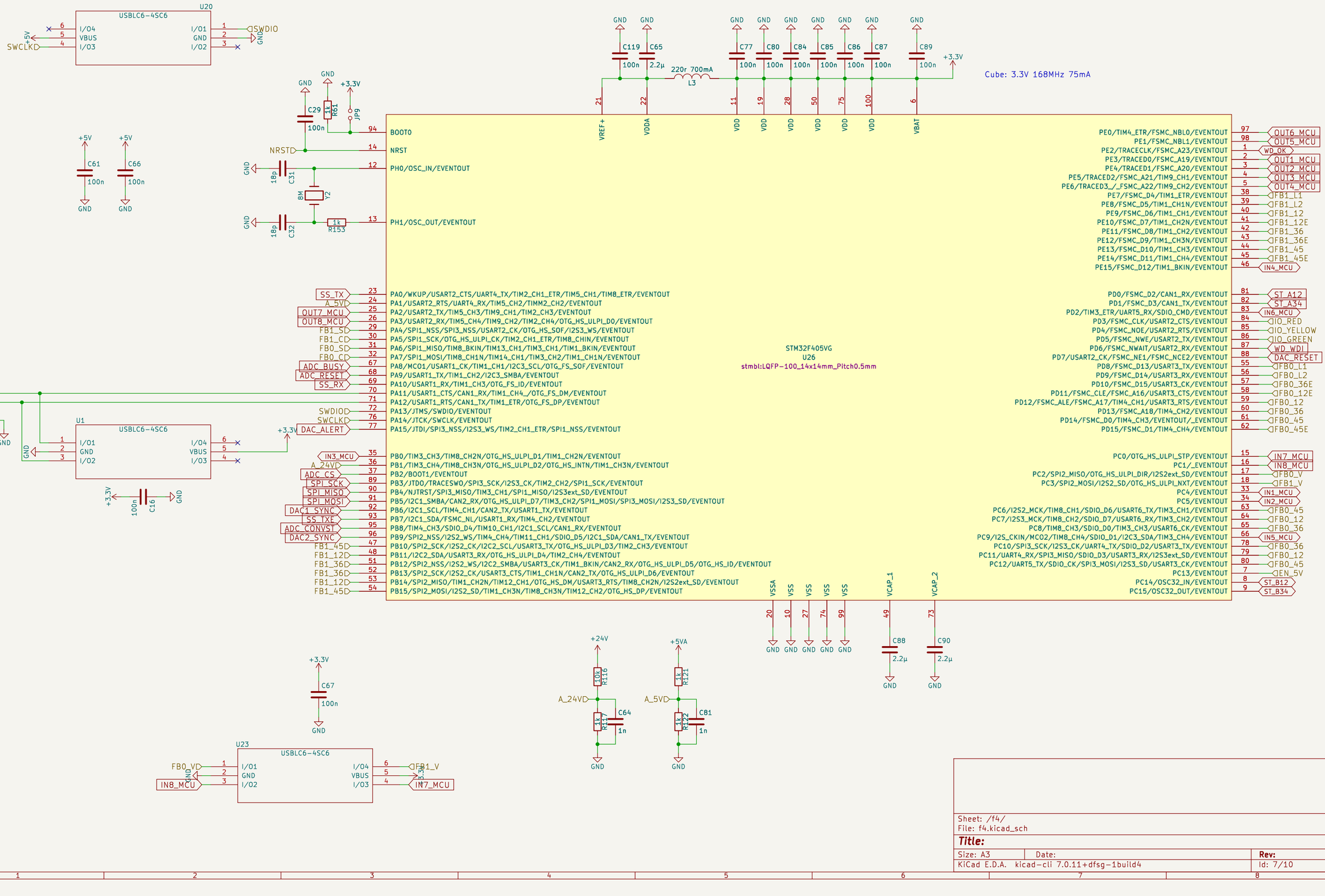
Sheet: /CMD/ File: cmd.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 3/10


SN65176B
U14
U17
U18
stmbl:SOIC-8-N
VCC
RO
-RE-
DE
DI
GND
A
B
D12
D36
D45
E12
E36
E45
C72
C74
C75
100n
120 R109
120 R111
120 R114
1k R123
1k R125
1k R128
220r 700mA
L9
C115
+5F
J2
RJ45_LED
L1
L2
470 R3
470 R6
C114
C125
10k R135
10k R137
JP2
JUMPER3_NC
+12V
A78
U6A
U6B
LM358
15k R47
10k R42
10k R44
C27
18p
15k R55
15k R50
10k R36
10k R39
C25
15k R56
22 R86
22 R89
C47
C48
470 R93
470 R95
1n
A12
A35
+5VA
220r 700mA
L8
100n C113
120 R130
120 R24
22 R147
120 R25
C13
100n
C36
LM358 max. output: 4.94V − 1.5V = 3.44V
STM32 max. input: 3.44V / (22 + 470) * 470 = 3.28V
4.94V / (120 + 120 + 120 + 22) * (120 + 22) = 1.83V
Sheet: /FB0/
File: fb.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4
Id: 4/10

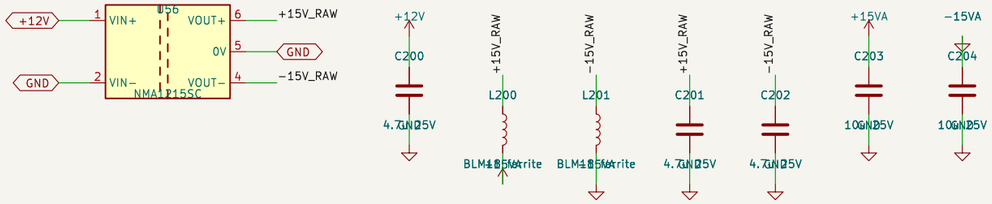

SN65176B
U15
U16
U19
stmbl:SOIC-8-N
VCC
RO
-RE-
DE
DI
GND
A
B
D12
D36
D45
E12
E36
E45
C71
C73
C76
100n
220r 700mA
L11
C118
+5F
L1
L2
470
R4
R5
RJ45_LED
J1
GNDD
C117
+12V
JP5
JUMPER3_NC
10k
R138
R140
A78
C126
1k
R139
15k
R48
R41
C26
18p
R43
R54
U7A
LM358
22
R87
C46
1n
470
R94
A12
LM358 max. output: 4.94V - 1.5V = 3.44V
STM32 max. input: 3.44V / (22 + 470) * 470 = 3.28V
+5VA
220r 700mA
L10
100n
C116
120
R131
R23
R148
R26
C12
C37
R49
R37
R38
C24
R57
U7B
R88
C49
R96
A35
4.94V / (120 + 120 + 120 + 22) * (120 + 22) = 1.83V
Sheet: /FB1/
File: fb.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4
Id: 5/10



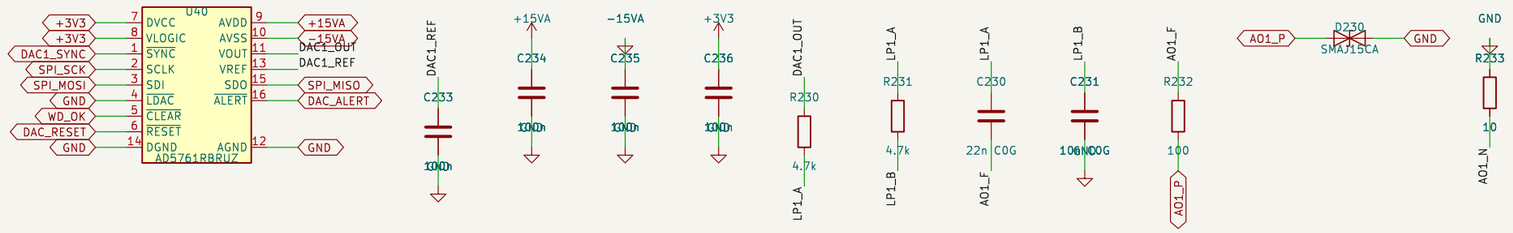
Sheet: /IO/		
File: io.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 6/10


U20
USBLC6-4SC6
I/O4
VBUS
I/O3
I/O1
GND
I/O2
SWCLK
SWDIO
+5V
GND
C61
C66
100n
C29
R61
1k
JP9
NRST
C31
C32
18p
8M
Y2
R153
BOOT0
NRST
PH0/OSC_IN/EVENTOUT
PH1/OSC_OUT/EVENTOUT
C119
C65
2.2µ
220r 700mA
L3
C77
C80
C84
C85
C86
C87
C89
+3.3V
Cube: 3.3V 168MHz 75mA
VREF+
VDDA
VDD
VBAT
STM32F405VG
U26
stmbl:LQFP-100_14x14mm_Pitch0.5mm
PA0/WKUP/USART2_CTS/UART4_TX/TIM2_CH1_ETR/TIM5_CH1/TIM8_ETR/EVENTOUT
PA1/USART2_RTS/UART4_RX/TIM5_CH2/TIMM2_CH2/EVENTOUT
PA2/USART2_TX/TIM5_CH3/TIM9_CH1/TIM2_CH3/EVENTOUT
PA3/USART2_RX/TIM5_CH4/TIM9_CH2/TIM2_CH4/OTG_HS_ULPI_D0/EVENTOUT
PA4/SPI1_NSS/SPI3_NSS/USART2_CK/OTG_HS_SOF/I2S3_WS/EVENTOUT
PA5/SPI1_SCK/OTG_HS_ULPI_CK/TIM2_CH1_ETR/TIM8_CHIN/EVENTOUT
PA6/SPI1_MISO/TIM8_BKIN/TIM13_CH1/TIM3_CH1/TIM1_BKIN/EVENTOUT
PA7/SPI1_MOSI/TIM8_CH1N/TIM14_CH1/TIM3_CH2/TIM1_CH1N/EVENTOUT
PA8/MCO1/USART1_CK/TIM1_CH1/I2C3_SCL/OTG_FS_SOF/EVENTOUT
PA9/USART1_TX/TIM1_CH2/I2C3_SMBA/EVENTOUT
PA10/USART1_RX/TIM1_CH3/OTG_FS_ID/EVENTOUT
PA11/USART1_CTS/CAN1_RX/TIM1_CH4_/OTG_FS_DM/EVENTOUT
PA12/USART1_RTS/CAN1_TX/TIM1_ETR/OTG_FS_DP/EVENTOUT
PA13/JTMS/SWDIO/EVENTOUT
PA14/JTCK/SWCLK/EVENTOUT
PA15/JTDI/SPI3_NSS/I2S3_WS/TIM2_CH1_ETR/SPI1_NSS/EVENTOUT
PB0/TIM3_CH3/TIM8_CH2N/OTG_HS_ULPI_D1/TIM1_CH2N/EVENTOUT
PB1/TIM3_CH4/TIM8_CH3N/OTG_HS_ULPI_D2/OTG_HS_INTN/TIM1_CH3N/EVENTOUT
PB2/BOOT1/EVENTOUT
PB3/JTDO/TRACESWO/SPI3_SCK/I2S3_CK/TIM2_CH2/SPI1_SCK/EVENTOUT
PB4/NJTRST/SPI3_MISO/TIM3_CH1/SPI1_MISO/I2S3ext_SD/EVENTOUT
PB5/I2C1_SMBA/CAN2_RX/OTG_HS_ULPI_D7/TIM3_CH2/SPI1_MOSI/SPI3_MOSI/I2S3_SD/EVENTOUT
PB6/I2C1_SCL/TIM4_CH1/CAN2_TX/USART1_TX/EVENTOUT
PB7/I2C1_SDA/FSMC_NL/USART1_RX/TIM4_CH2/EVENTOUT
PB8/TIM4_CH3/SDIO_D4/TIM10_CH1/I2C1_SCL/CAN1_RX/EVENTOUT
PB9/SPI2_NSS/I2S2_WS/TIM4_CH4/TIM11_CH1/SDIO_D5/I2C1_SDA/CAN1_TX/EVENTOUT
PB10/SPI2_SCK/I2S2_CK/I2C2_SCL/USART3_TX/OTG_HS_ULPI_D3/TIM2_CH3/EVENTOUT
PB11/I2C2_SDA/USART3_RX/OTG_HS_ULPI_D4/TIM2_CH4/EVENTOUT
PB12/SPI2_NSS/I2S2_WS/I2C2_SMBA/USART3_CK/TIM1_BKIN/CAN2_RX/OTG_HS_ULPI_D5/OTG_HS_ID/EVENTOUT
PB13/SPI2_SCK/I2S2_CK/USART3_CTS/TIM1_CH1N/CAN2_TX/OTG_HS_ULPI_D6/EVENTOUT
PB14/SPI2_MISO/TIM1_CH2N/TIM12_CH1/OTG_HS_DM/USART3_RTS/TIM8_CH2N/I2S2ext_SD/EVENTOUT
PB15/SPI2_MOSI/I2S2_SD/TIM1_CH3N/TIM8_CH3N/TIM12_CH2/OTG_HS_DP/EVENTOUT
PC0/OTG_HS_ULPI_STP/EVENTOUT
PC1/_EVENTOUT
PC2/SPI2_MISO/OTG_HS_ULPI_DIR/I2S2ext_SD/EVENTOUT
PC3/SPI2_MOSI/I2S2_SD/OTG_HS_ULPI_NXT/EVENTOUT
PC4/EVENTOUT
PC5/EVENTOUT
PC6/I2S2_MCK/TIM8_CH1/SDIO_D6/USART6_TX/TIM3_CH1/EVENTOUT
PC7/I2S3_MCK/TIM8_CH2/SDIO_D7/USART6_RX/TIM3_CH2/EVENTOUT
PC8/TIM8_CH3/SDIO_D0/TIM3_CH3/USART6_CK/EVENTOUT
PC9/I2S_CKIN/MCO2/TIM8_CH4/SDIO_D1/I2C3_SDA/TIM3_CH4/EVENTOUT
PC10/SPI3_SCK/I2S3_CK/UART4_TX/SDIO_D2/USART3_TX/EVENTOUT
PC11/UART4_RX/SPI3_MISO/SDIO_D3/USART3_RX/I2S3ext_SD/EVENTOUT
PC12/UART5_TX/SDIO_CK/SPI3_MOSI/I2S3_SD/USART3_CK/EVENTOUT
PC13/EVENTOUT
PC14/OSC32_IN/EVENTOUT
PC15/OSC32_OUT/EVENTOUT
PD0/FSMC_D2/CAN1_RX/EVENTOUT
PD1/FSMC_D3/CAN1_TX/EVENTOUT
PD2/TIM3_ETR/UART5_RX/SDIO_CMD/EVENTOUT
PD3/FSMC_CLK/USART2_CTS/EVENTOUT
PD4/FSMC_NOE/USART2_RTS/EVENTOUT
PD5/FSMC_NWE/USART2_TX/EVENTOUT
PD6/FSMC_NWAIT/USART2_RX/EVENTOUT
PD7/USART2_CK/FSMC_NE1/FSMC_NCE2/EVENTOUT
PD8/FSMC_D13/USART3_TX/EVENTOUT
PD9/FSMC_D14/USART3_RX/EVENTOUT
PD10/FSMC_D15/USART3_CK/EVENTOUT
PD11/FSMC_CLE/FSMC_A16/USART3_CTS/EVENTOUT
PD12/FSMC_ALE/FSMC_A17/TIM4_CH1/USART3_RTS/EVENTOUT
PD13/FSMC_A18/TIM4_CH2/EVENTOUT
PD14/FSMC_D0/TIM4_CH3/EVENTOUT/_EVENTOUT
PD15/FSMC_D1/TIM4_CH4/EVENTOUT
PE0/TIM4_ETR/FSMC_NBL0/EVENTOUT
PE1/FSMC_NBL1/EVENTOUT
PE2/TRACECLK/FSMC_A23/EVENTOUT
PE3/TRACED0/FSMC_A19/EVENTOUT
PE4/TRACED1/FSMC_A20/EVENTOUT
PE5/TRACED2/FSMC_A21/TIM9_CH1/EVENTOUT
PE6/TRACED3_/_FSMC_A22/TIM9_CH2/EVENTOUT
PE7/FSMC_D4/TIM1_ETR/EVENTOUT
PE8/FSMC_D5/TIM1_CH1N/EVENTOUT
PE9/FSMC_D6/TIM1_CH1/EVENTOUT
PE10/FSMC_D7/TIM1_CH2N/EVENTOUT
PE11/FSMC_D8/TIM1_CH2/EVENTOUT
PE12/FSMC_D9/TIM1_CH3N/EVENTOUT
PE13/FSMC_D10/TIM1_CH3/EVENTOUT
PE14/FSMC_D11/TIM1_CH4/EVENTOUT
PE15/FSMC_D12/TIM1_BKIN/EVENTOUT
VSSA
VSS
VCAP_1
VCAP_2
C88
C90
2.2µ
SS_TX
A_5V
OUT7_MCU
OUT8_MCU
FB1_SD
FB1_CD
FB0_SD
FB0_CD
ADC_BUSY
ADC_RESET
SS_RX
SWDIO
SWCLK
DAC_ALERT
IN3_MCU
A_24V
ADC_CS
SPI_SCK
SPI_MISO
SPI_MOSI
DAC1_SYNC
SS_TXE
ADC_CONVST
DAC2_SYNC
FB1_45
FB1_12
FB1_36
FB1_36
FB1_12
FB1_45
OUT6_MCU
OUT5_MCU
WD_OK
OUT1_MCU
OUT2_MCU
OUT3_MCU
OUT4_MCU
FB1_L1
FB1_L2
FB1_12
FB1_12E
FB1_36
FB1_36E
FB1_45
FB1_45E
IN4_MCU
ST_A12
ST_A34
IN6_MCU
IO_RED
IO_YELLOW
IO_GREEN
WD_WDI
DAC_RESET
FB0_L1
FB0_L2
FB0_36E
FB0_12E
FB0_12
FB0_36
FB0_45
FB0_45E
IN7_MCU
IN8_MCU
FB0_V
FB1_V
IN1_MCU
IN2_MCU
FB0_45
FB0_12
FB0_36
IN5_MCU
FB0_36
FB0_12
FB0_45
EN_5V
ST_B12
ST_B34
USB_OTG
VCC
D-
D+
ID
GND
shield
P1
C1
100n
120
R1
U1
USBLC6-4SC6
C16
U23
FB0_V
IN8_MCU
FB1_V
IN7_MCU
C67
+24V
+5VA
A_24V
A_5V
10Ω
R116
1k
R117
C64
1n
R121
R122
C81
Sheet: /f4/
File: f4.kicad_sch
Title:
Size: A3
Date:
Rev:
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4
Id: 7/10

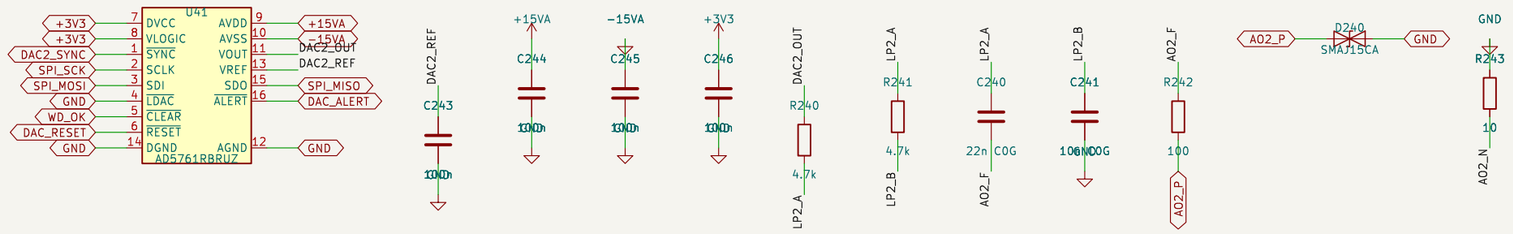
±15 V: NMA1215 from +12 V, LC filtered



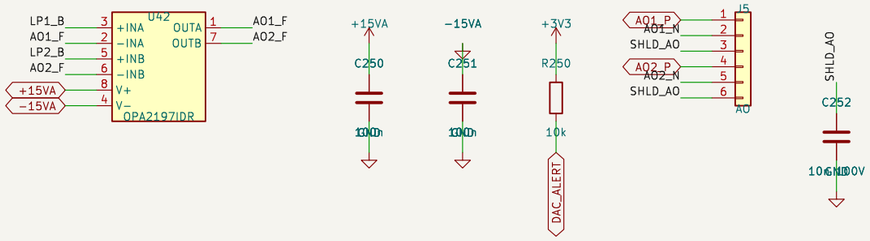
AO1: DAC, Sallen–Key (R230 R231 C230 C231, U42A), 100 R, TVS



AO2 (fitted for the spindle's C axis): DAC, Sallen–Key (U42B)



Filter op–amp, /ALERT, connector J5



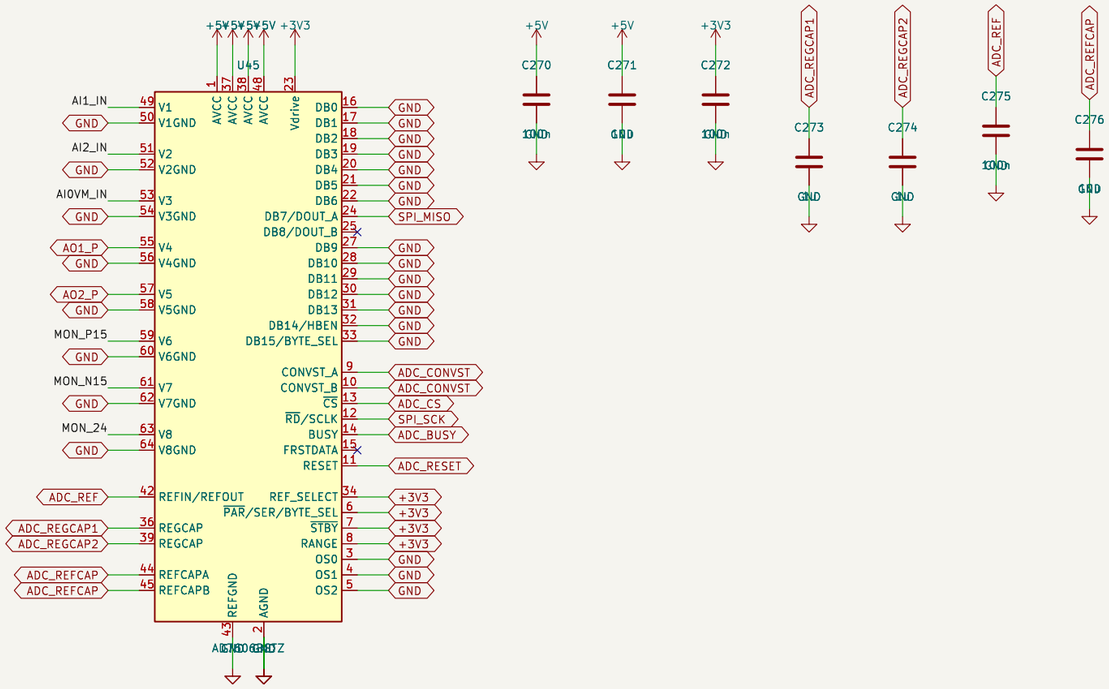
* in a part's Note: check against the datasheet
docs/stman.md

Sheet: /dac/
File: dac.kicad_sch

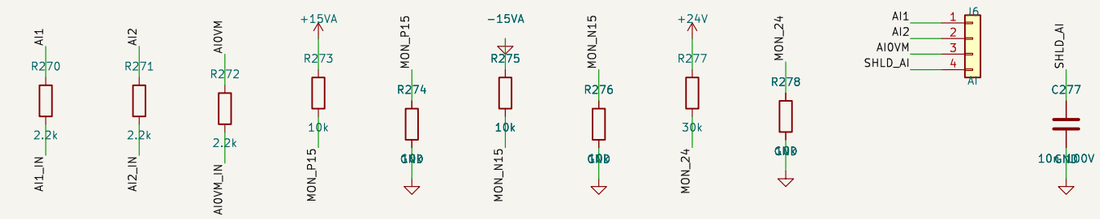
Title: stman: Command outputs AO1, AO2: AD5761R, 2nd–order low–pass 2.3

Size: A3	Date:	**Rev:**
KiCad E.D.A. kicad–cli 7.0.11+dfsg–1build4		Id: 10/10

AD7606: 8 ch simultaneous, ±10 V (RANGE high), serial (PAR/SER high), internal reference



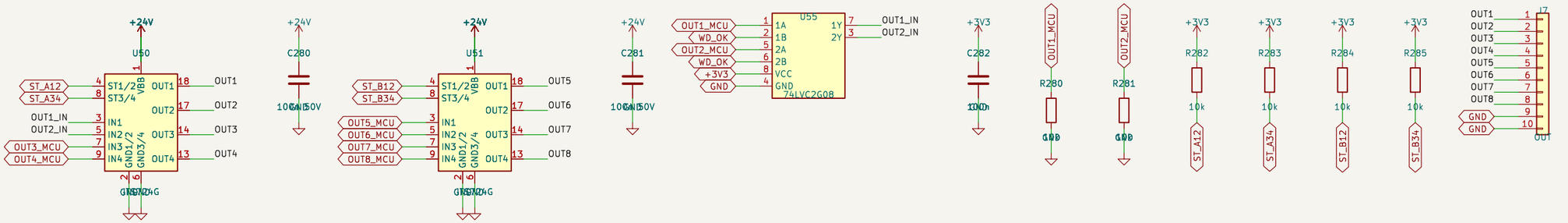
Inputs: 2.2k series (the ADC's 1 MΩ inputs, ±16.5 V clamps); rail dividers; J6



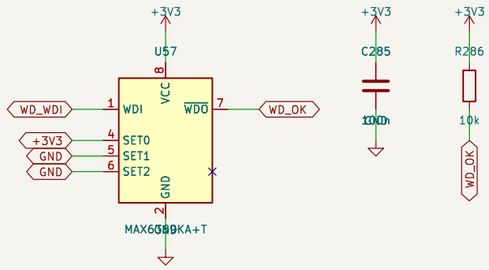
* in a part's Note: check against the datasheet
docs/stman.md

Sheet: /ain/		
File: ain.kicad_sch		
Title: stman: Monitor ADC: AD7606 (drive monitors and 0VM, AO readbacks, ra		
Size: A3	Date:	**Rev:**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 11/10

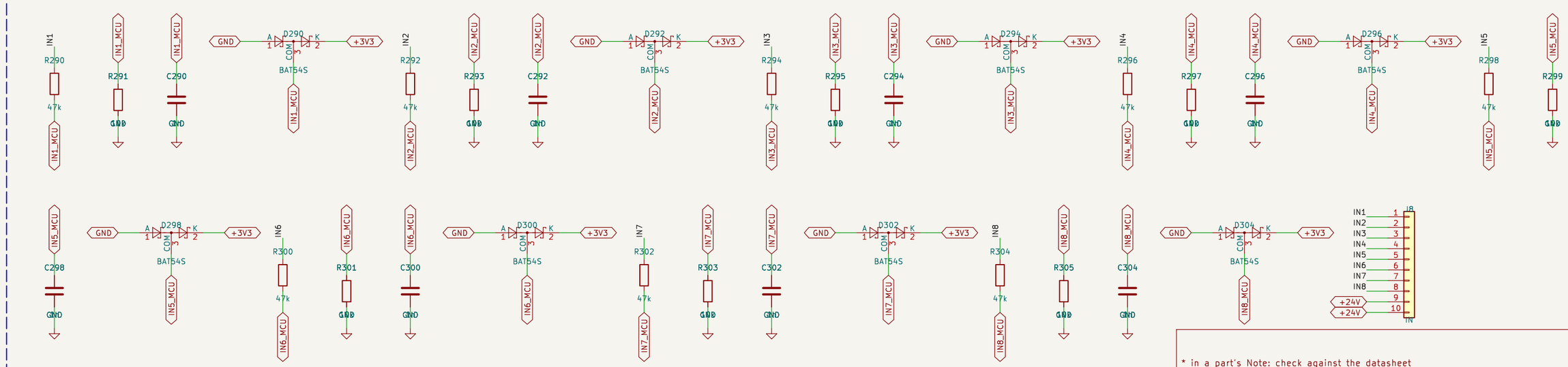
Outputs: ITS724G x2, OUT1/OUT2 gated by the watchdog



Watchdog MAX6369: WD_OK low -> DACs to 0 V, OUT1/OUT2 off



Inputs: 47k / 10k, 1 nF, BAT54S clamp (high > ~13 V, low < ~6 V)



* in a part's Note: check against the datasheet docs/stman.md

Sheet: /drvio/
File: drvio.kicad_sch

Title: stman: Drive I/O: 8 x 24 V out (ITS724G), 8 x 24 V in, watchdog

Size: A3	Date:	Rev:
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 12/10